

Heaven's light is our guide

Rajshahi University of Engineering & Technology



Electrical and Computer Engineering

Lab report-03

Course code: ECE -2112

Course title: Digital Techniques Sessional

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Lab-03: Logic circuit diagram simulation from Boolean expression

Theory

A Boolean expression is used to describe a logic function with Boolean variables and operations such as **AND**, **OR**, and **NOT**. These expressions can be represented using logic gates to create a digital logic circuit.

In this experiment, the given Boolean expression is converted into a logic circuit using basic gates. The **AND gate** performs the AND operation, the **OR gate** combines the required terms, and the **NOT gate** produces the complemented variables. The outputs of these gates are connected according to the given expression.

For example, consider the Boolean expression:

$$F = AB + \bar{B}C + AC + \bar{A}B\bar{D}$$

Here, the AND gates are used to generate the individual product terms, while the OR gate combines them to produce the final output F . NOT gates are used wherever complemented variables such as $\bar{A}$, $\bar{B}$, and $\bar{D}$ are required.

Logisim Evolution can be used to build and test the logic circuit. Different input combinations can be applied as **0 or 1**, and the resulting output can be observed. This allows the circuit to be checked against the given Boolean expression and truth table, helping to confirm that the circuit operates correctly.

Task-01: $F = AB + \bar{B}C + AC + \bar{A}B\bar{D}$

Simplification:

AB\CD	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	1	1	1
10	0	1	1	0

So,

$$F = AB + \bar{B}C + BD$$

Verification with Logisim Evolution:

Output:

Format:

Style:

Notation:

		C, D			
		00	01	11	10
A, B	00	0	0	1	1
	01	1	0	0	1
	11	1	1	1	1
	10	0	0	1	1

No group selected.

$$\overline{B} \cdot C + B \cdot \overline{D} + A \cdot B$$

Figure 1: Logisim verification for task-01

Implement with logic gate:

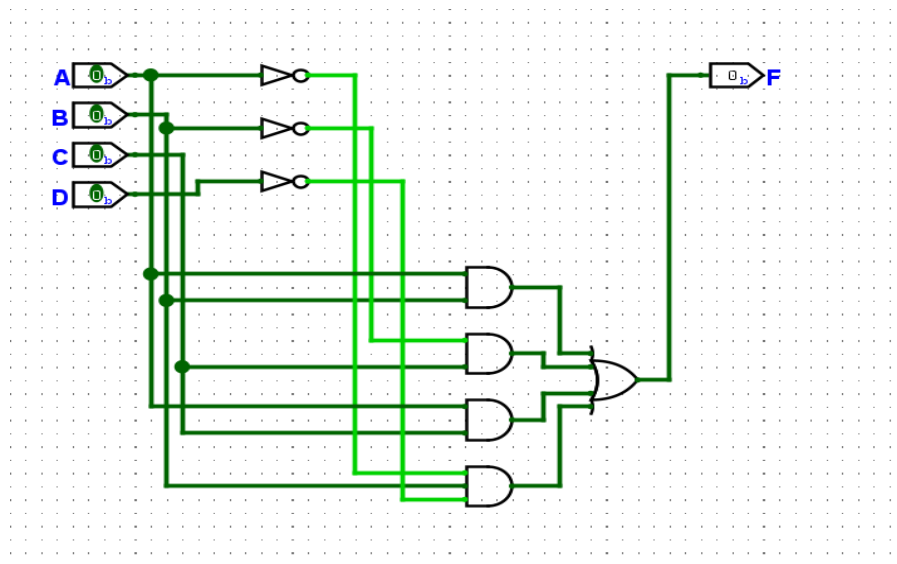


Figure 2: Logic gate implementation for task-01

Task-02: $F = \overline{(\bar{A} + B)(C + \bar{D})} [\bar{A}B + (\bar{C}\bar{D})]$

Simplification

Using De Morgan's theorem:

$$\overline{(\bar{A} + B)(C + \bar{D})} = \overline{(\bar{A} + B)} + \overline{(C + \bar{D})} = A\bar{B} + \bar{C}D$$

Also,

$$\overline{(\bar{C}\bar{D})} = C + \bar{D}$$

Therefore,

$$F = (A\bar{B} + \bar{C}D)(\bar{A}B + C + \bar{D})$$

Using the Boolean identity,

$$(X + Y)(X + Z) = X + YZ$$

we get,

$$F = A\bar{B} + \bar{C}\bar{D}(C + \bar{D})$$

Now,

$$\bar{C}\bar{D}(C + \bar{D}) = \bar{C}\bar{D}C + \bar{C}\bar{D}\bar{D} = 0 + 0 = 0$$

Therefore,

$$F = A\bar{B}$$

Verification with Logisim Evolution:

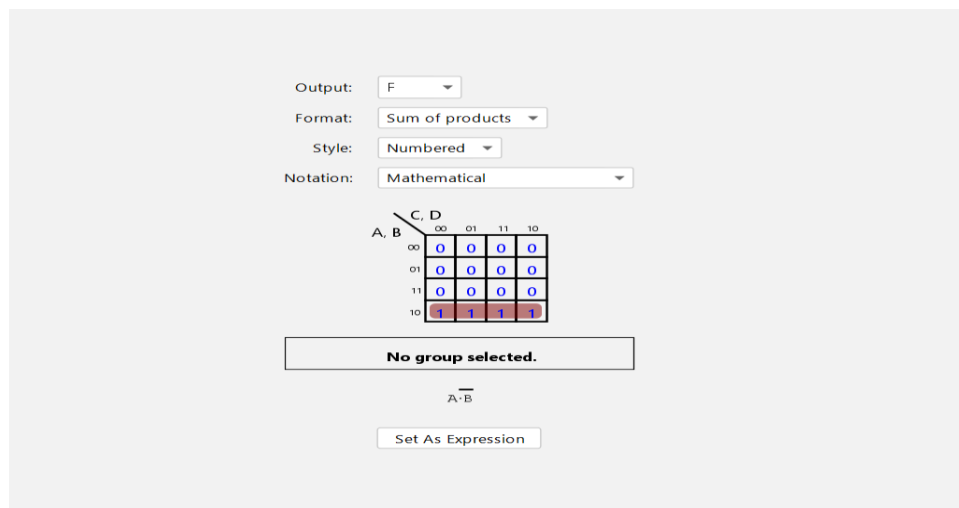


Figure 3: Logisim verification for task-02

Implement with logic gate:

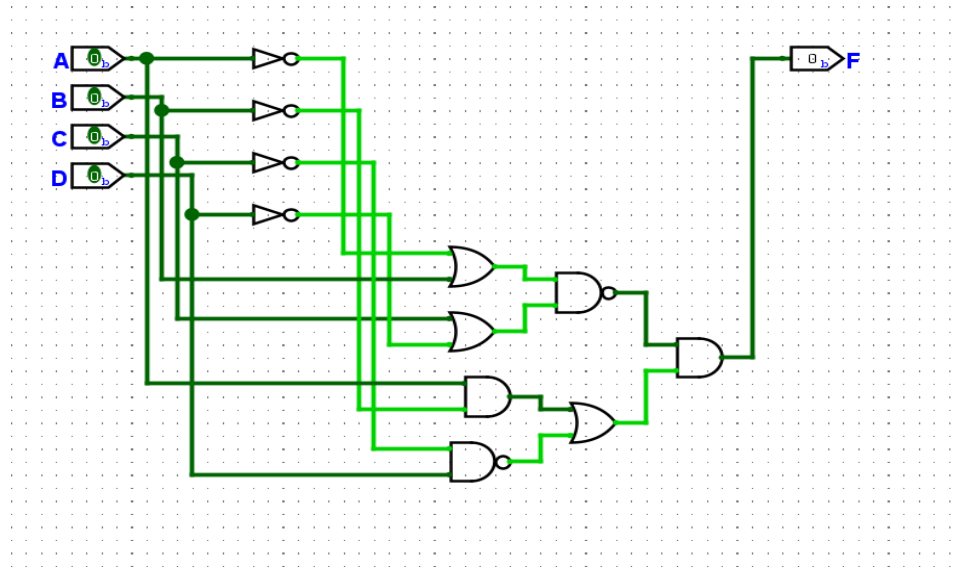


Figure 4: Logic gate implementation for task-02

Task-03: $F = ABCD + \overline{A}\overline{B}CD + \overline{A}B\overline{C}D + A\overline{B}\overline{C}D + ABC\overline{D}$

Simplification:

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	0

So,

$$F = ABC + ABD + ACD + BCD$$

Verification using Logisim Evolution:

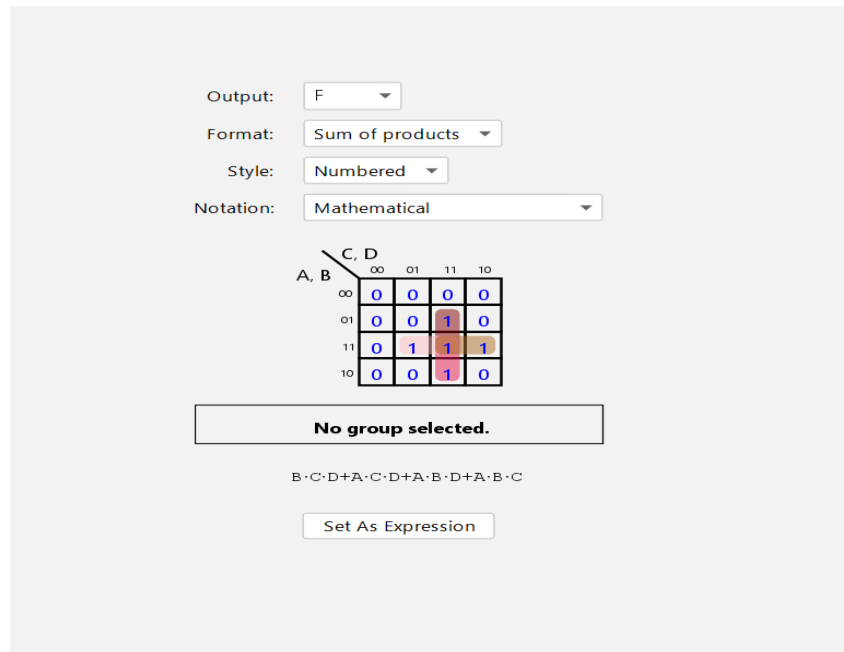


Figure 5: Logisim verification for task-03

Implement with logic gate:

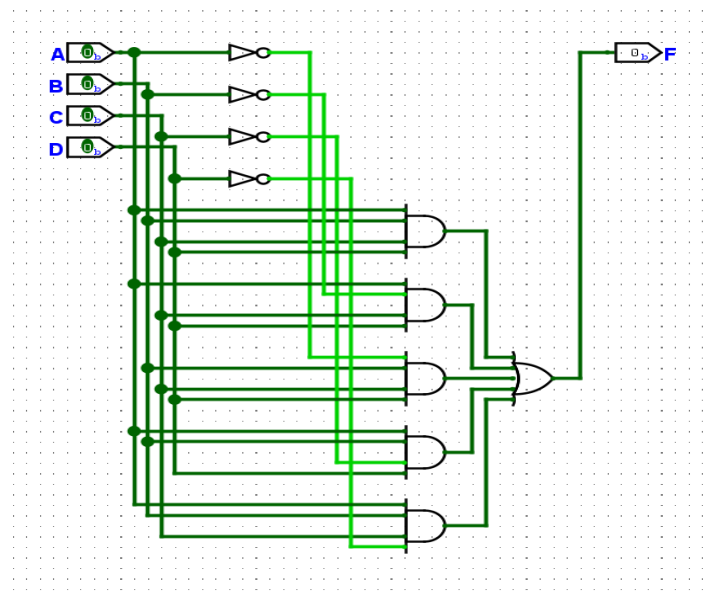


Figure c: Logic gate implementation for task-03

Task-04: $F = (A + B + C + D)(A + B + \bar{C})(A + \bar{B} + C)(\bar{A} + B + C)$

Simplification:

AB\CD	00	01	11	10
00	0	1	0	0
01	0	0	1	1
11	1	1	1	1
10	0	0	1	1

So,

$$F = AB + AC + BC + \bar{A}\bar{B}\bar{C}\bar{D}$$

Verification with Logisim Evolution:

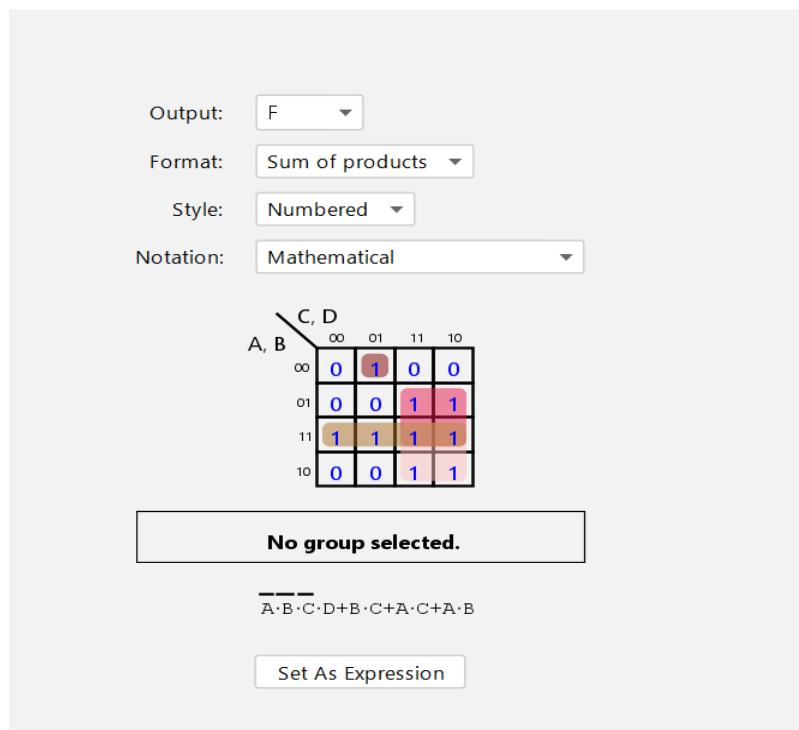


Figure 7: logisim Verification for task-04

Logic Gate Implementation:

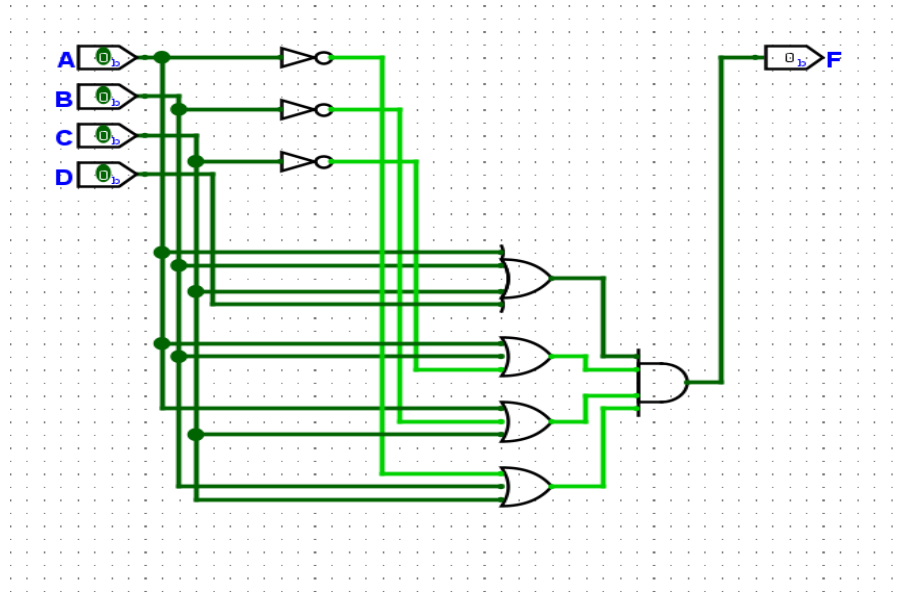


Figure 8: Logic gate implementation for task-04

Task-05: $F = [A + \overline{B}(C + D)][\overline{A} + B(\overline{C} + \overline{D})]$

Simplification:

$$F = [A + \overline{B}(C + D)][\overline{A} + B(\overline{C} + \overline{D})]$$

$$F = \overline{A}A + AB(\overline{C} + \overline{D}) + \overline{A}\overline{B}(C + D) + \overline{B}B(C + D)(\overline{C} + \overline{D})$$

$$F = AB(\overline{C} + \overline{D}) + \overline{A}\overline{B}(C + D)$$

$$F = ABC + ABD + \overline{A}\overline{B}C + \overline{A}\overline{B}D$$

So,

$$F = ABC + ABD + \overline{A}\overline{B}C + \overline{A}\overline{B}D$$

Verification with Logisim Evolution:

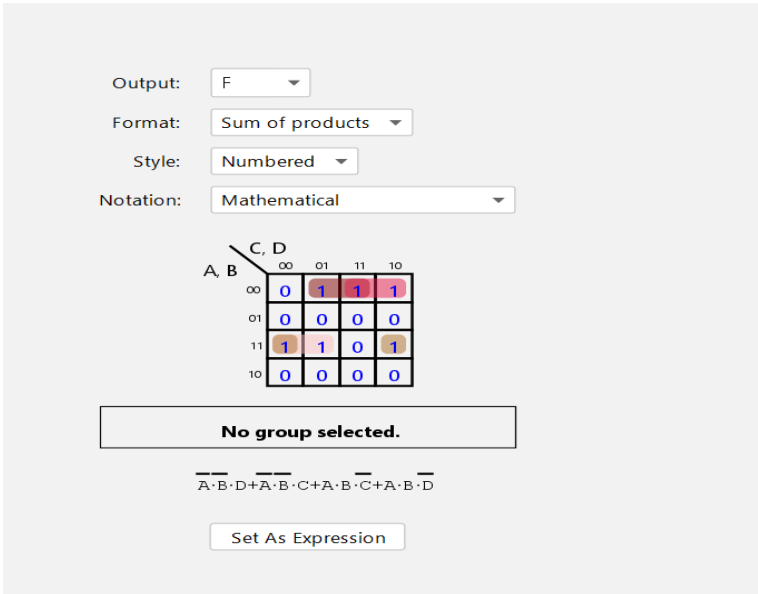


Figure S: Logisim Verification for task-05

Logic Gate Implementation:

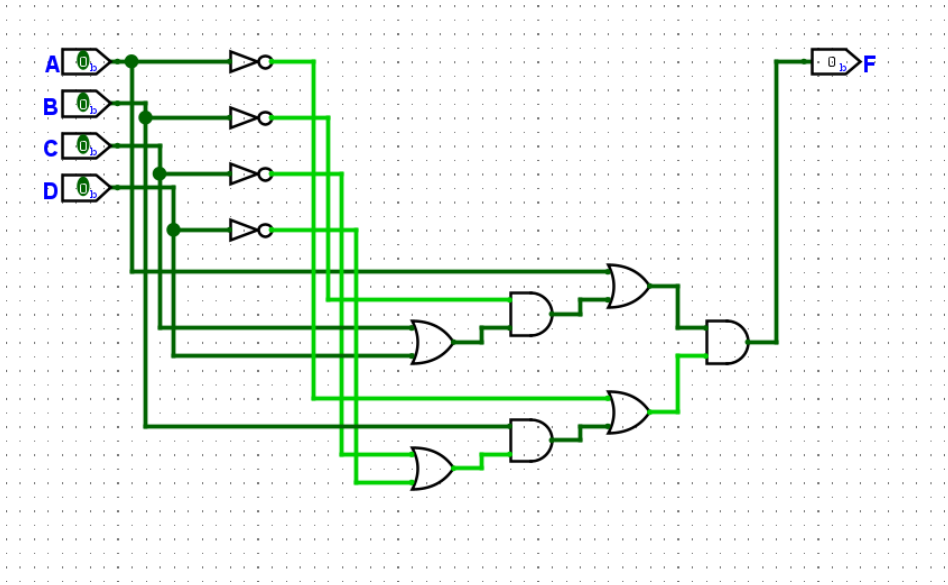


Figure 10: Logic gate implementation for task-05

Discussion:

In this experiment, the given Boolean expressions were simplified and implemented using **Logisim Evolution**. Boolean algebra and **Karnaugh maps (K-maps)** were used to simplify the expressions, and the resulting circuits were built using **AND, OR, and NOT gates**.

The simplified expressions required fewer terms and helped make the circuits easier to construct. Different input combinations were then applied in Logisim Evolution to check the outputs. The obtained results matched the expected Boolean functions, confirming that the circuits were implemented correctly.

Overall, the experiment provided a practical understanding of **Boolean expressions, K-map simplification, logic gates, and digital circuit simulation**. It also showed how simplification techniques can be used to make logic circuits more efficient.